

Study Guide

Electromagnetics I (5EPF0)

**Department of Electrical Engineering
Electromagnetics Group**

**Academic year 2026-2027
Document version: August 5, 2026**

1 Introduction

This study guide explains the goals and procedures for the BSc course Electromagnetics I (5EPF0). It serves in addition to the regular information that is available through Osiris. The most up-to-date information is always available through Canvas.

2 General information

Course title:	Electromagnetics I
Course code:	5EPF0
ECTS credits:	5
Quartile:	1
Timeslot:	D (Wed 5–8; Fri 1–4)
Program:	Bachelor College EE and AU, year 2, compulsory Bachelor College SI – Sustainable Energy (EN), year 2, compulsory Pre-master program EE, compulsory

Type of education: *LECTURES:*

Lectures are 4 hours per week. Lectures will be recorded through the Yuja service, through which they are served online. A series of lectures for the predecessor of this course (5EPA0) and earlier years of this course are also available through this service. Please note that the content may differ.

Lectures will be held at the time-slots **13.30–15.15 on Wednesdays** and **8.45–10.30 on Fridays**.

Questions about the lectures, or the course in general are discussed during the lectures, but can also be asked to the responsible lecturers (Sander Bronckers and Bart Smolders) during the tutorials, where they will walk around.

The general schedule is depicted in Section 5.

TUTORIALS:

Instructions will take place on campus.

Each on-campus room will have instructors present at all times. During the instructions, we (lecturers) will do our best to attend all the rooms in turns as well. This will give you the opportunity to ask questions about the lectures or general course proceedings.

If you have questions, about the instruction-problems, including the exam-level problems, you can ask them during the tutorials. Questions about the instruction-problems will be addressed via hints. All questions are welcome, so also questions you may have about earlier or later instruction-problems. For the quizzes, help will only be available after all the quizzes have been handed in, since points are awarded for it.

Instructions will be held at the time-slots **15.30–17.15 on Wednesdays** and **10.45–12.30 on Fridays**.

Course subject:	This course gives a first introduction into electromagnetism. Many electrical engineering concepts can be derived from electromagnetism. Electromagnetic fields and applications are all around us; just think of microwaves, the antenna of your cell phone, electric machines, traffic radar or an MR scanner in a hospital.
Lecturers:	Sander Bronckers (responsible lecturer) Bart Smolders
Course material:	<i>Book:</i> W.H. Hayt and J.A. Buck, Engineering Electromagnetics, McGraw-Hill International Edition, 8th edition (H&B8), 2012, or 9th edition (H&B9), 2017, Available via Thor <i>Other:</i> Study guide, lecture slides, and pencasts. Made available on Canvas (canvas.tue.nl).
Assessment:	Exam: 3-hour (written) exam in AnsDelft format. Exams are on-campus, in written form. Only basic (non-graphic) calculators are allowed during the exam. On-campus quizzes: There will be 13 on-paper, on-campus ANS quizzes that take place during the last half hour of the instructions of weeks 1 through 7 (week 8 is reserved for self-study). <i>Quizzes can only be done on-paper during the dedicated timeslots in the instruction rooms.</i> Collaboration is allowed, but the use of AI (see section 8) is not. Only the best nine out of thirteen results of the quizzes will be counted. This will give ample of opportunity for students to compensate for bad results due to absence or technical difficulties. Therefore, we will not be open to discuss these issues unless there are very special circumstances. The exam and the quizzes count for 90% and 10%, respectively. To pass the course the mark for the exam should be equal or in excess of 5.0, and the overall grade should be equal or in excess of 5.5.
Professional skills:	None
Group:	Electromagnetics (EM), Department of Electrical Engineering

3 Learning objectives and content

At the end of the course, a student will/should be able to:

- describe in his/her own words the relevance of electromagnetism in the various applications within Electrical Engineering and Automotive, e.g. wireless communication, medical applications, and electromechanics;
- apply mathematical concepts and skills related to vector calculus in various coordinate systems to electromagnetic problems;
- apply and interpret physical concepts from electrostatics (Coulomb, Gauss), magnetostatics (Biot-Savart, Ampère, Lorentz) and electrodynamics (fields and waves, Faraday, Maxwell, Poynting);
- define the mathematical models of these physical concepts in various coordinate systems, both in differential and integral form, where appropriate;
- apply the mathematical models for the design of electrical components, such as capacitors, inductors, transformers, and basic electric engines;

- apply the finite difference method as a numerical method to solve electrostatic problems and interpret simulation results.

The contents of the course can be grouped into three main topics:

- Electrostatics
 - Electrostatic fields, Coulomb, Superposition, Gauss's law
 - Work and electric (scalar) potential
 - Capacitance, dielectric materials
 - Laplace and Poisson equations
 - Energy
 - Finite difference method for electrostatics (Poisson, Laplace)
- Magnetostatics
 - Rotation (curl), Stokes' theorem
 - Current, current density, magnetic flux density and Ampère's law
 - Vector potential and Biot-Savart
 - Lorentz force
 - Magnetic fields and magnetic materials
 - Inductance (self, mutual)
 - Boundary conditions
- Time-dependent fields and waves
 - Faraday
 - Lenz's law
 - Continuity equation, displacement current
 - Maxwell's equations in differential and integral form
 - Maxwell's equations in frequency domain, phasors
 - One-dimensional waves, plane waves (time domain and time harmonic), free space impedance
 - Skin effect

4 Position in the curriculum

Assumed prior knowledge:	Mathematics 1 and 2 Physics for EE, or Physics for AT
Follow-up courses:	Electromechanics Electromagnetics II

5 Course schedule

In the schedule below, you can find the dates and locations of the live lectures and tutorials. Roughly, the material is divided into seven weeks of two blocks of lectures, plenary questions, and tutorials.

	Week 1 2/Sep	Week 2 9/Sep	Week 3 16/Sep	Week 4 23/Sep	Week 5 30/Sep	Week 6 7/Oct	Week 7 14/Oct	Week 8 21/Oct	
Wed	1								
	2								
	3								
	4								
	5	LEC1	LEC3	LEC5	LEC7	LEC8	LEC10	LEC12	Lectures
	6							ET	
	7	INS1	INS3	INS5	INS7	INS8	INS10	INS12	Tutorials
	8							ET	Quizes

	4/Sep	11/Sep	18/Sep	25/Sep	2/Oct	9/Oct	16/Oct	23/Oct	
Fri	1	LEC2	LEC4	LEC6	Momen- Tum	LEC9	LEC11	LEC13	Lectures
	2								
	3	INS2	INS4	INS6	no Quiz	INS9	INS11	INS13	Tutorials
	4							ET	Quizes
	5								
	6								
	7								
	8								

LEC Lecture

INS Instruction

ET Exam Training

The final written exam is scheduled on November 5th 2026, 13:30-16:30.

The re-sit is scheduled on January 28th 2027, 18:00-21:00.

6 Education format and assessment

Education format

For this course, 8 hours per week are planned in Q1. There are 4 hours of lectures per week, accompanied by 4 hours of tutorials per week.

The detailed contents of the tutorials can be found on Canvas.

Assessment

The course Electromagnetics I (5EPF0) consists of the following partial grades:

- Final written test, accounting for 90% of your grade;
- On-campus ANS tests twice a week (best 9 of 13 count), accounting for 10% of your grade;

Remarks:

- At the final test, you will be provided with a formula sheet. These will also be made available through Canvas, so you can get used to them. Next to a ruler, pen and paper, and basic (non-graphic) calculator, the EM1 formula sheet (available on canvas) is the only material that you are allowed to use during the final test;
- You have to score 5.0 or higher for the final written test (or its' re-sit) to pass the course;

7 Course subjects per week

In the table below, the overview of the material taught in the course is given per lecture. The schedule is tentative and may still be subject to change.

Lecture	Subject matter	Book Section
1	Introduction	
	Re-cap vector calculus, vector fields	1.1 – 1.7
	Re-cap coordinate systems	1.8, 1.9
	Coulomb's law	2.1
2	Superposition	2.2
	Electrostatic fields	2.3 – 2.5
	Electric flux density	3.1
	Gauss's law	3.2 – 3.4
3	Gauss's law continued	3.2 – 3.4
	Re-cap divergence	3.5, 3.6
	Work and energy	4.1
	Re-cap line integral	4.2
4	Electric potential	4.3 – 4.5
	Re-cap gradient	4.6
	Energy density	4.8
	Current, current density	5.1, 5.2
	Metallic conductors	5.3
5	Dielectric materials	5.6, 5.7
	Electric boundary conditions	5.4, 5.8
	Capacitance	6.1 – 6.3
6	Poisson's and Laplace's equation	6.6, 6.7
	Finite difference techniques for electrostatics	Additional material
	Electrostatics 'cookbook'	Formula sheet
7	Magnetic field intensity	7.1
	Biot & Savart	7.1
	Ampère's law	7.2
8	Re-cap curl / cross-product	7.3
	Stokes' theorem	7.4
	Magnetic flux density	7.5
9	Magnetic vector potential	7.6
	Lorentz force	8.1 – 8.4
	Magnetic materials	8.6
10	Magnetic boundary conditions	8.7
	Faraday's law	9.1
	Magnetostatics 'cookbook'	Formula sheet
11	Inductance	8.10
	Potential energy	8.9
	Maxwell's equations	9.3, 9.4

12	Displacement current	9.2
	Electromagnetic waves	11.1
13	Wave propagation in materials	11.2
	Skin effect	11.4
	Electromagnetics 'cookbook'	Formula sheet
ET	Exam training	

8 Academic integrity

8.1 Ground rules for students

Be aware that you should always operate with academic integrity, taking into account the rules and regulations about copyright, plagiarism and fraud. Use the links below to learn more about what you are (not) allowed to do in your assignments and assessments:

- [TU/e Rules for AI use in Education](#)
- [TU/e Regulations and Code of Conduct](#)
- [TU/e Copyright Information Point](#)
- [EE BSc Regulations and PER](#)
- [EE MSc Regulations and PER](#)
- [AI & ES MSc Regulations and PER](#)

8.2 On the use of AI

You are not allowed to use generative AI for any of the assessment forms of this course, including the quizzes. Quizzes can only be done on-paper during the dedicated timeslots in the instruction rooms

Furthermore, we strongly advise you NOT to use AI for any of the instruction problems or exam preparation unless specifically asked for, as (1) they may not suggest a correct approach and (2) you will need to be able to solve these problems independently, which is significantly more difficult than understanding a pre-made analysis of the problem. It is for this same reason that the full solutions to the exercises are only made available after their corresponding instructions. If you are stuck on a problem, we are there to help you!